

QCC Labs Hosting a Lab

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Goal: Allow you to host a lab for you and your friends

One of the goals I had for the Interactive Labs is that I wanted to host free training for our attendees. I wanted to take someone with little to no experience in a topic, help them build an environment to learn in, teach them a bit, and then at the end of the 2 hour session the learners could take your lab home with you.

The first year of the Interactive Labs room (QCC 0x2, 2024), I placed myself in a very interesting place. 5 weeks out we had no labs so I took ownership of the room on Friday afternoon. At that moment I started building out the first 7 labs. I had some simpler one and ones that took a bit more time. I reached out to my peers and got people that would volunteer to be trainers for QCC. I allowed them to select the lab that they wanted (fist come first server) and trained the trainer. As year 1 was a bit rushed I resorted to purchasing cheap thumb drives to load files on. I did not like this solution long term because now I had lost the integrity of each flash drive as I put them down on the tables. What I wanted to avoid a a merry prankster slipping malware or inappropriate content (such as cat / kitten photos #TeamPuppy).

After the conference I spent a lot of time figuring out what I wanted to do. I had an idea of a portable travel router that I could hook a large SDD to it. After much research I chose the glinet GL-AXT1800. At the time this was one of the top reviewed travel routers, had advanced functionality (that I did not need), was small enough that I could fit it in my pocket, could host file sharing, and ran off USB-C (I like to try to standardize on single open power charging solutions. Flash forward to QCC 0x3, 2025 I showed up with a smile on my face and ran into some challenges:

- While the device could do fileshares, it could not do more well known file shares. Either people would need to install a webdav client or manually map a SMB share. Please keep in mind I want to spend as MUCH time as possible on the training and helping the attendees learn and less time on them getting the files.
- While my home speed tests were acceptable, they broke down with too many people the wifi trying to xfer files.

So I had two challenges that I needed to overcome:

- Increase the speed of the file transfers. (SIGNIFICANTLY)
- Make it as easy as possible for anyone to download the files.
- Keep the build to something I could take to QCC and other cons.

I feel the need, the need for speed!

So the bottleneck I was having was that my travel router wifi was a BIGGLY bottleneck! Back when I worked as a network engineer I always promote the mantra 'be the packet' during troubleshooting. Starting at the OSI model layer 1 and walking up. It occurred to me that I could take a significant burden off my travel router I could increase speed.

- Wanting to balance speed, portability, and quietness I purchased an 8 port 10 / 1 gig managed switch. You see, the way the data link layer (layer 2 of the OSI model) is very different than the way roting (layer 3 of the OSI model works. Think of it this way your street acts like a VLAN, or all of the devices on your internal network. In many ways it functions like a street, in than you can go directly out to your street and directly access any of the other houses on your street. However if you want to go to a house on another street you must go through an intersection, or router that will connect one or more networks to each other. We us a lightweight service on the travel router to hand out IP addresses to Interactive Labs clients and host wifi. In total I really only want 6 ports live to keep speeds up for the attendees. I got a special USB-C 10/1 Ethernet converter for the lab host, and four USB-C 10/1 Ethernet for learners to barrow.
- Now I needed to get the file share off the travel router! To do this I went back to my island of misfit toys. I was gifted an old 2-in-one enterprise style windows tablet a few years ago. Whilethe specs are pretty good, the detachable keyboard fails to work about half the time. What are you going to do so I wiped it and put my favorite Linux on it. I lost things that I did not care for in a lab machine such as auto rotation of screen and touch interface. I am sure I could get both those working if I cared to, but I don't. I did not want to be in a place where I could not run the lap due to the kyboard not working so I purchased a Bluetooth foldable keyboard which is very nice and solves almost all of the problem except in this version of linux you MUST decrypt the drive before Bluetooth is enabled so I wiped the machine and at install purposely choose NOT to encrypt the drive. I know, best practices and all but this system is ONLY going to be used to host lab files so I do not see an issue.

Now I have the QCC Lab server, I have the network setup. What I need now is an easy way to host the files from the Lab Laptop. My mind went instantly wanting to be able to host it through HTTP / HTTPS. Trueth be told, I found a lot of apps that would host in HTTP. The downfall was one of the of the major browsers no longer allos HTTP site access, EVEN if you put in the approved 'work-a-round' in their active! HOW RUNE!!!

So I went back hunting until I found https://localsend.org/.

> "
>
> Open source and cross-platform file sharing for everyone.
>
> "

It is a small light weight app I can use to share files for QCC, has multi-platform support, and users can either install the app or assess the HTTPS website. It is basic but, so am I! So lets get started. Also this is now a default app I install on all of my systems because it just works.

The first things I did after I opened the app was I went into setting and expanded all settings. Then I:

- Network -> Device name I set this to something more appropriate for each
- Network -> Default Port (Local send by default uses TCP 53317, who wants to be basic #NOT ME)

On windows changing the port can cause the app to break intermittently, to fix the error:

> SocketException: Failed to create server socket (OS Error: Only one usage of each socket address (protocol/network address/port) is normally permitted.
> , errno = 10048), address = 0.0.0.0, port = 1337

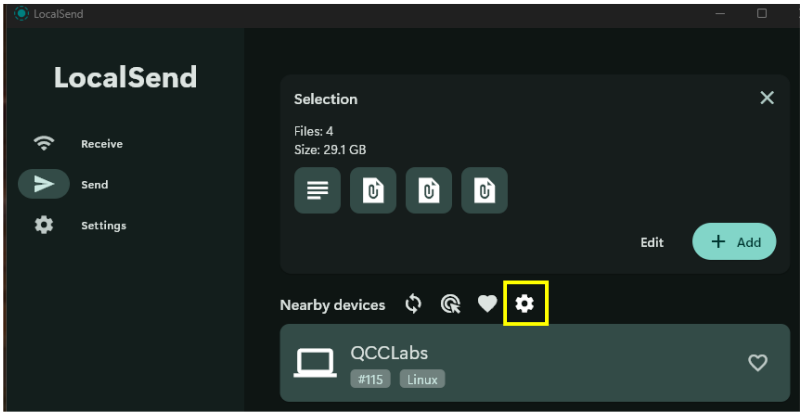
netstat -ano | findstr :[port]
taskkill /PID <PID_NUMBER> /F

from an admin command line run the netstat command with the port number that you wish to use substituted for [port].

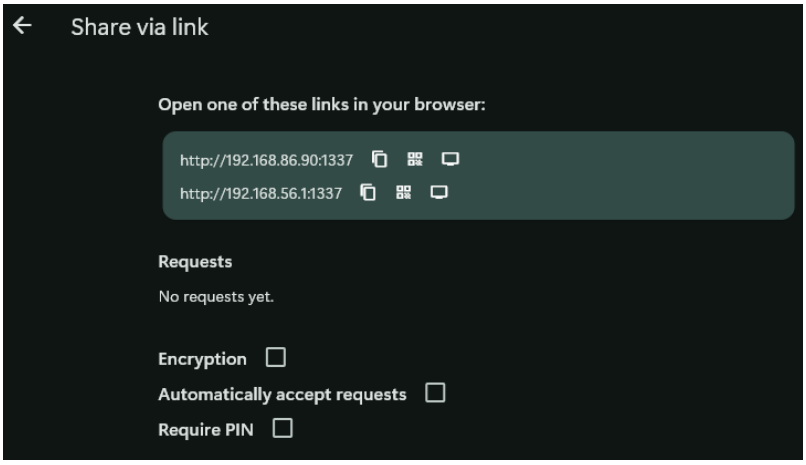
Then run the taskkill command with the returned port id number from the netstat command in <PID_number>. I am not sure you need to do this but I reboot after just to make things clean.

OK now all the steps that you need to do to set you lab share up the way I do:

1. Click 'Send' -> 'Folder'
    a. Select the folder you wish to share and click 'Select folder'
2. In the right half of the screen click the gears icon just to the right of 'Nearby Devices'

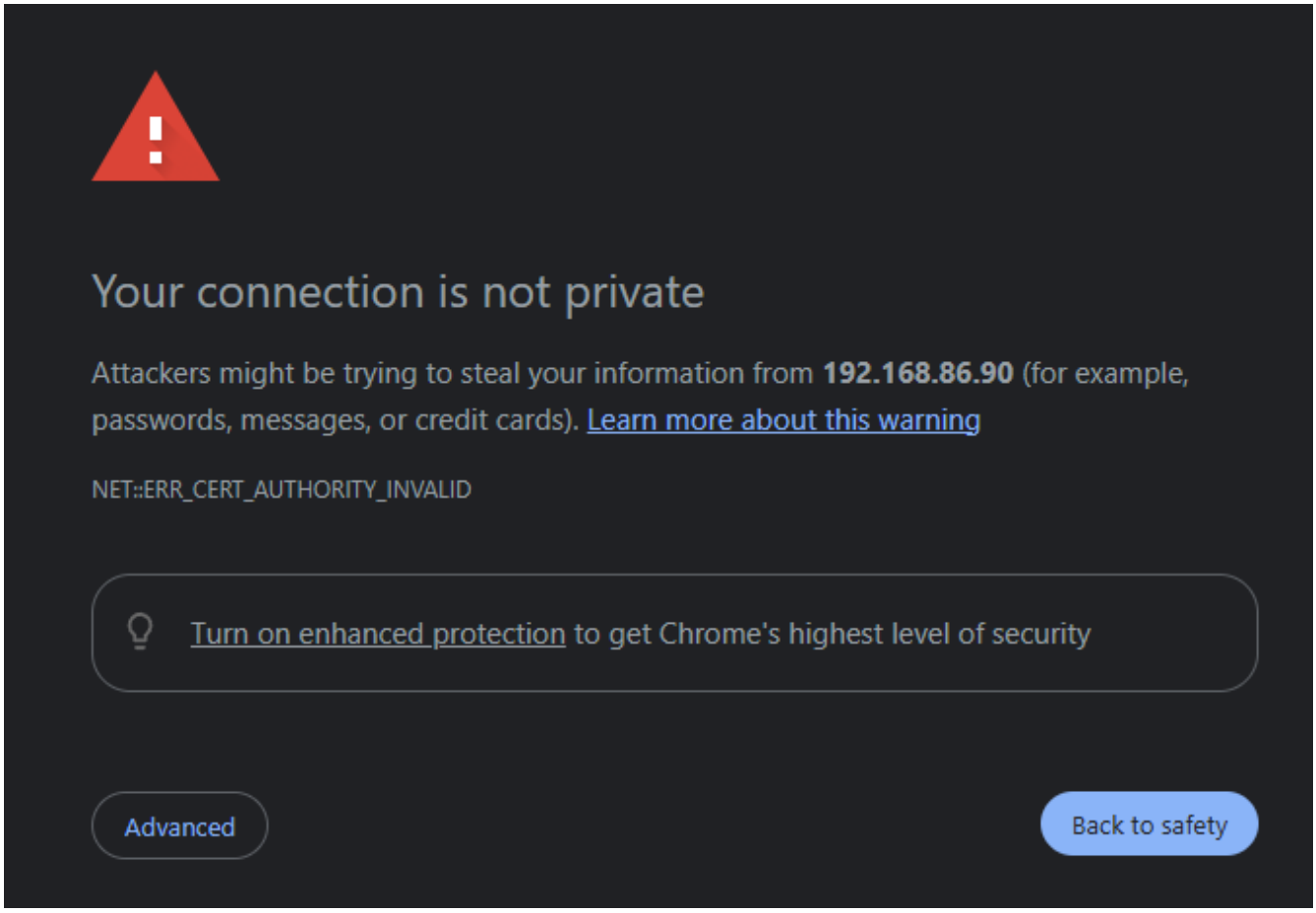


3. Click 'Share Via Link'



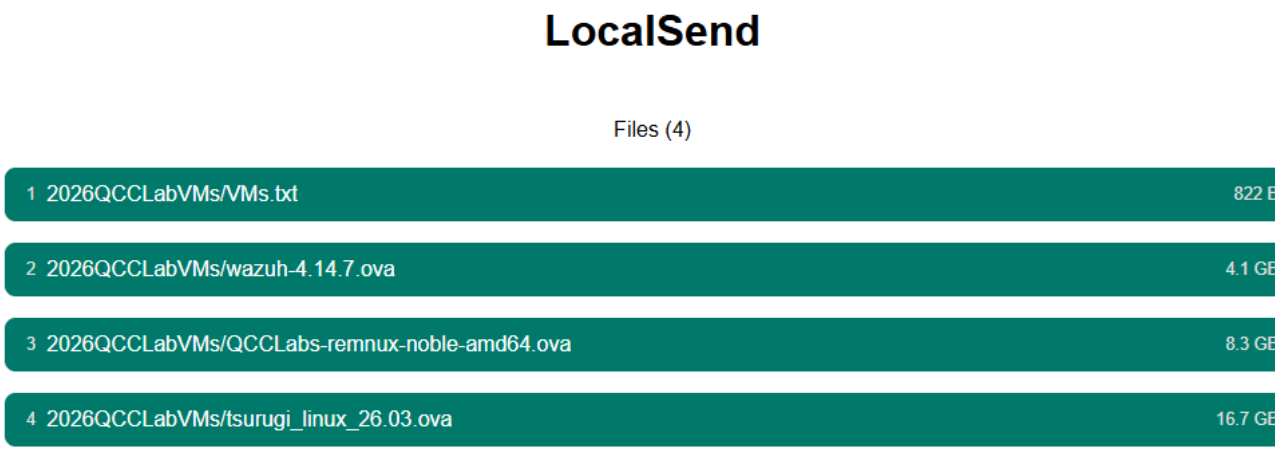
4. Click both 'Encryption' and 'Automatically accept requests'.
5. In this case I have two networks
    a. 192.168.86.90 is my actual IP on my home network
    b. 192.168.56.1 is the local VLAN from my virtual box instance.

6. Now attended can open the browser of their choice and go to https://192.168.86.90:1337



   a. Based on an attenees browser they will get a slightly different message. This is due to LocalSend using a perfectly valid self signed certificate. Basically we need to tell the browser 'Yes, I really want to go here'. In my case this is by clisking 'Advanced' -> 'Proceed to 192.168.86.90 (unsafe). It is flagged as unsafe ONLY because we are using a self signed cert! #HowRude

   b. Success



The advantages of using this setup

- As long as my lab server is on and the screen is locked, no one can make changes to the files I am serving.
- Attendees can access all lab files through the web browser already on their system!
- Setup of the actual share takes a minute so we can spend more time with the students.
- We ask if people just want the ZIP of all the lab content, please feel free to grab it via wifi, if people want the larger files, please leverage the ethernet switch based connection (balance ease of use and speed)

If you have a network connection on your host machine and set you VM host to a 'Bridged Adapter' then your systems will be on the same network. This will allow you to use LocalSend on one device to quickly send files to the other system!

Until Next time!